M3 Screw
GND
M3 Screw
GND

Electrovalvula
File: Electrovalvula.kicad_sch
ELECTROVALVULA

Buffer
File: Buffer.kicad_sch
BUFF_ENABLE
Pump_Signal1
Pump_Signal2
Pump_Signal3
Pump_Signal1CN
Pump_Signal2CN
Pump_Signal3CN

Power Supply
File: Power_Supply.kicad_sch

MCU
File: MCU.kicad_sch
ELECTROVALVULAD
BUFF_ENABLED
Pump_Signal1D
Pump_Signal2D
Pump_Signal3D
BOMBA_1
BOMBA_2
BOMBA_3

Bomba_Refri
File: Bomba_Refri.kicad_sch
BOMBA_1
BOMBA_2
BOMBA_3
Pump_Signal1CN
Pump_Signal2CN
Pump_Signal3CN

Ethernet_PHY
File: Ethernet_PHY.kicad_sch
LAN_INT
RMII{RMII}
PHY_RST
LAN_INT
RMII{RMII}
PHY_RST

Caudalimetro
File: Caudalimetro.kicad_sch
CAUDALIMETRO{CAUDALIMETRO}
CAUDALIMETRO{CAUDALIMETRO}

SD
File: SD.kicad_sch
SDMMC{SDMMC}
SDMMC{SDMMC}

Transductores_Presion
File: Transductores_Presion.kicad_sch
PRESSURE_SENS{PRESSURE_SENS}
PRESSURE_SENS{PRESSURE_SENS}

CAN
File: CAN.kicad_sch
SLNT
CAN{CAN_Harness}
SLNT
CAN{CAN_Harness}

Regulador_Presion
File: Regulador_Presion.kicad_sch
REGULATOR_IN
REGULATOR_OUT
REGULATOR_IND
REGULATOR_OUT

Info_LEDs
File: Info_LEDs.kicad_sch
LEDS_INFO{LEDS_INFO}
LEDS_INFO{LEDS_INFO}

SDC
File: SDC.kicad_sch
SDC_CLOSED
SDC_CLOSED

RELEASED
BRAKE_FAULT
BRAKE_RESET

Braking_Sensor
File: Braking_Sensor.kicad_sch
BRAKE_RESET
BRAKE_FAULT
RELEASED
SENSORS{SENSORS}

Reed_Module
File: Reed_Module.kicad_sch
SENSORS{SENSORS}

PCB Manufacturing Notes

This PCB contains high speed analog and digital signals that should be routed with specific impedance profiles, using coplanar microstrip, stripline and differential routing techniques.

Details such as the specific impedance profile, trace width and trace spacing can be found in the layout notes in each sub-sheet of this PCB.

The impedance profiles are included in the Layer Stack Manager of this project and will only be accurate if the following aspects are specified when ordering:

Layers = 6
Inner copper weight = 0.5oz
Outer copper weight = 0.5oz + plating
PCB thickness = 1.6mm
Manufacturer = NCAB
Stack-up Code = 6ML BOARD – 1,6MM

You can find the details of the stackup 6ML BOARD – 1,6MM:

Contents	Thickness (mm)
Solder mask	0,02
0,5oz + plating	0,035
PP	0,200
CCL 0,43 1oz/1oz	0,035
	0,360
	0,035
PP	0,240
CCL 0,43 1oz/1oz	0,035
	0,360
	0,035
PP	0,200
0,5oz + plating	0,035
Solder mask	0,02

M3 Screw
GND
M3 Screw
GND

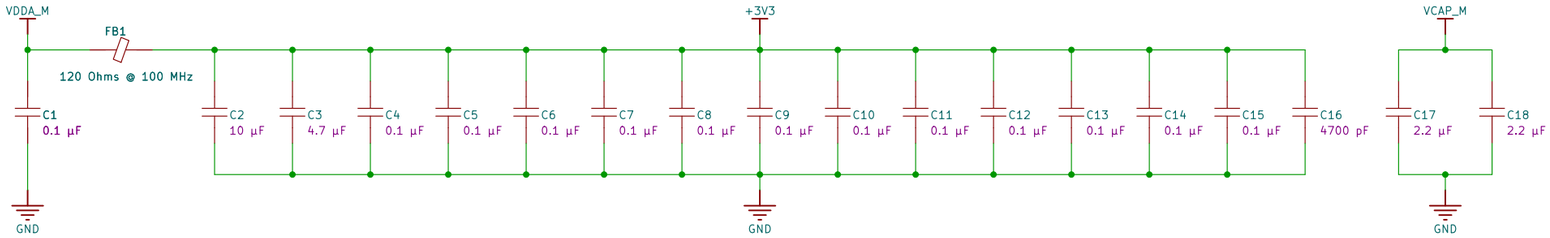
Sheet: /
File: VCU nueva nueva.kicad_sch

Title:

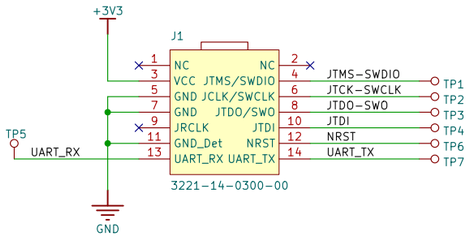
Size: A3 Number: ${DOCUMENTNUMBER} Date: ${CREATIONDATE}
Last Modification: ${MODIFIEDDATE} Sheet 1 of 16
Author: ${AUTHOR}
Checked by: ${CHECKEDBY}

HYPERLOOP UPV
Altium

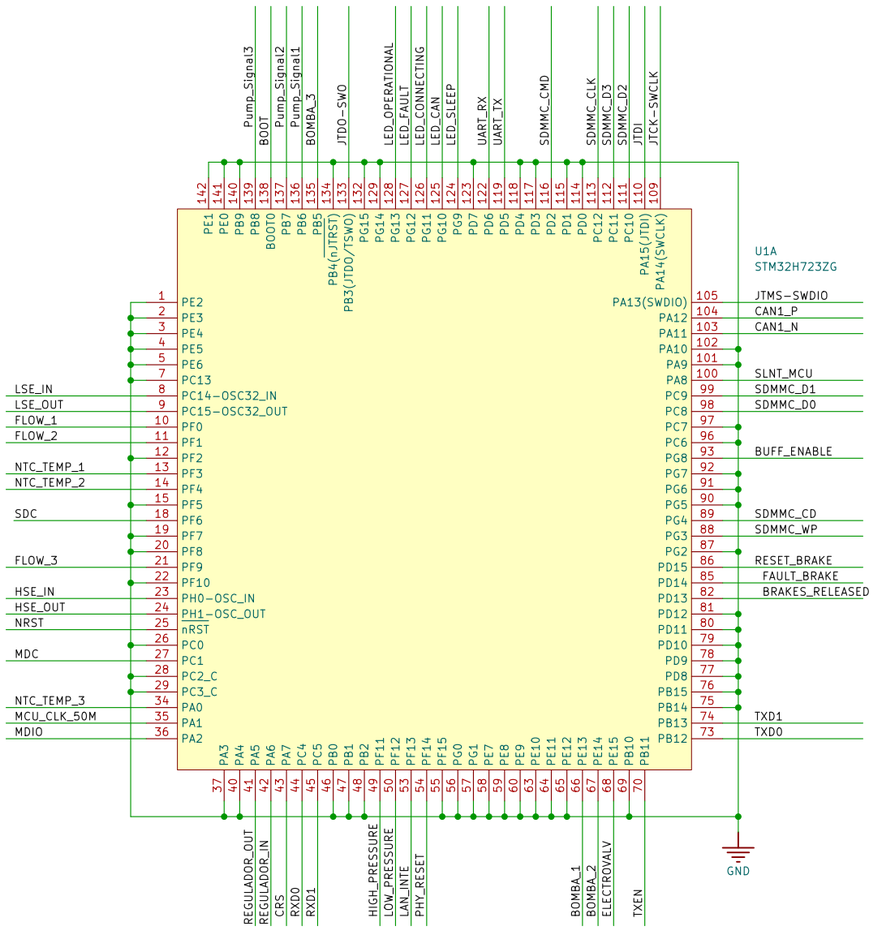
MCU Supply Decoupling



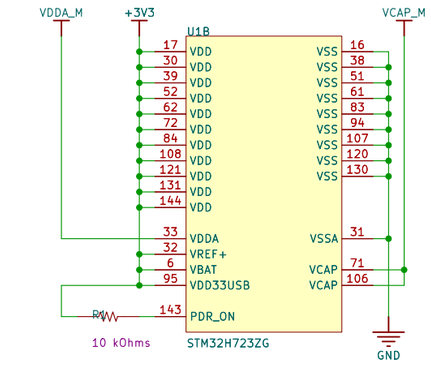
JTAG Interface



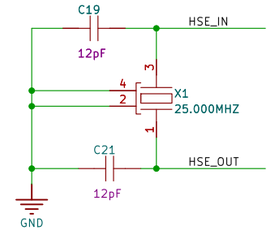
MCU



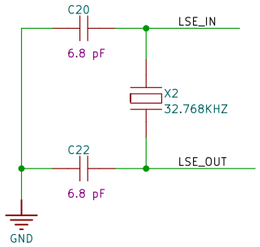
MCU Supply



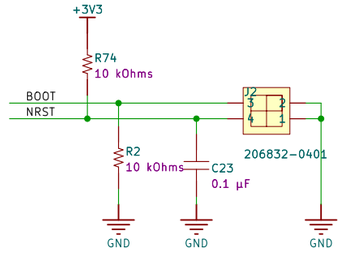
HSE Crystal



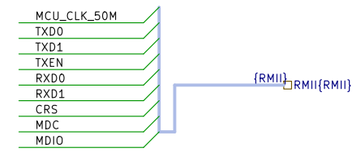
LSE Crystal

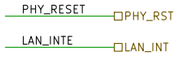


Boot

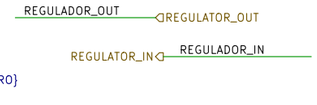


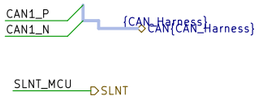


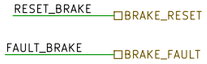






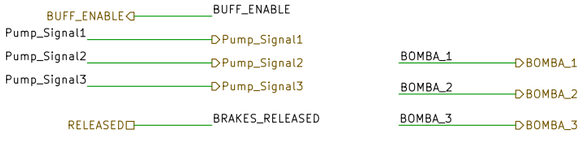








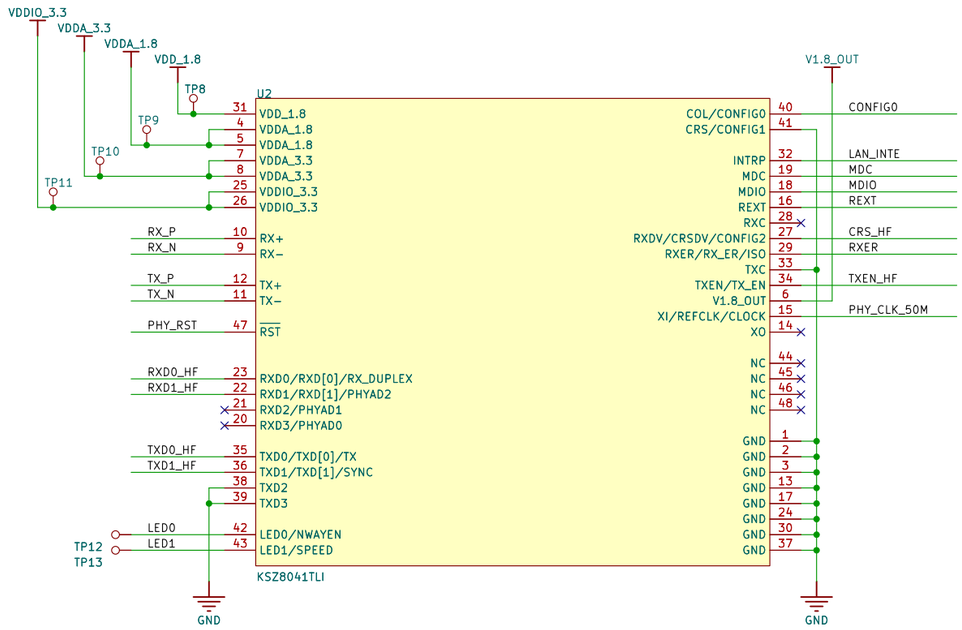






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Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 2/16

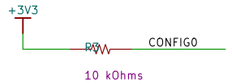
Ethernet Transceiver



Strap-In Options

Default Strap-ins:
PHY Adress = 00001
Isolate Mode = Disable
Speed = 100 Mbps
Auto-Negotiation = Enable
Interface = RMII

Pullup = RMII Interface
NC = MII Interface



Pulldown = Full-Duplex
NC = Half-Duplex

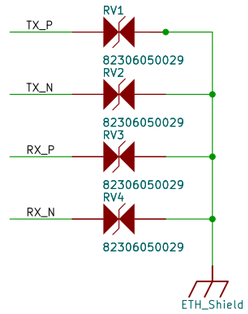


Create a ground plane beneath the ethernet connector with the ETH_Shield net.

Terminate ESD protection diodes to ETH_Shield plane.

Bond ETH_Shield to GND using ferrite beads at a single bonding point.

ESD Protection



Keep all RMII traces below 150mm.

Avoid using vias on RMII signals, if necessary, use the same number of vias for all signals.

Length match the following signals to <5 mm:

TXD0, TXD1, TXEN, length matched to each other (RMII_TX net class).

RXD0, RXD1, CRS, length matched to each other (RMII_RX net class).

PHY_CLK_50M, MCU_CLK_50M, length matched to each other (RMII_SYNC net class).

Layout Notes

All RMII signals need to be routed using a 50 Ohm coplanar microstrip or stripline impedance profile:

The calculations for trace width have been made according to the manufacturing specs listed in the main architecture sheet of this PCB and assuming a coplanar routing approach (0.1mm clearance to GND).

Note that the signal layers of this PCB are Layer 1, 4 and 6 and the reference layers are Layer 2 and 5. If this is changed, the trace width has to be recalculated.

The results of the JLCPCB online impedance caluclator are the following:

Layer 1 trace width = 0.1199 mm
Layer 4 trace width = 0.2149 mm
Layer 6 trace width = 0.1199 mm

The impedance profiles will only be accurate if the following aspects are specified when ordering:

Layers = 6
Inner copper weight = 0.5oz
Outer copper weight = 1oz
PCB thickness = 1.6mm
Manufacturer = JLCPCB
Stack-up Code = JLC0616H-3313A

Route analog RX (RX_P, RX_N) and TX (TX_P, TX_N) pairs as differential signals with a 100 Ohm impedance profile, in a coplanar differential microstrip or stripline configuration.

The calculations for trace width have been made according to the manufacturing specs listed in the main architecture sheet of this PCB and assuming a coplanar differential routing approach (0.1mm clearance to GND. 0.2mm distance between differential signal taces).

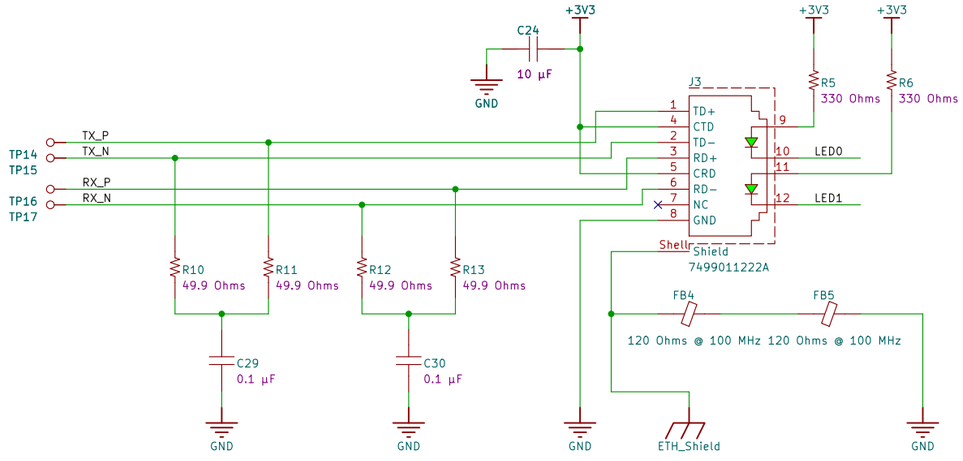
The results of the JLCPCB online impedance calculator are the following:

Layer 1 trace width = 0.1090 mm
Layer 4 trace width = 0.1476 mm
Layer 5 trace width = 0.1090 mm

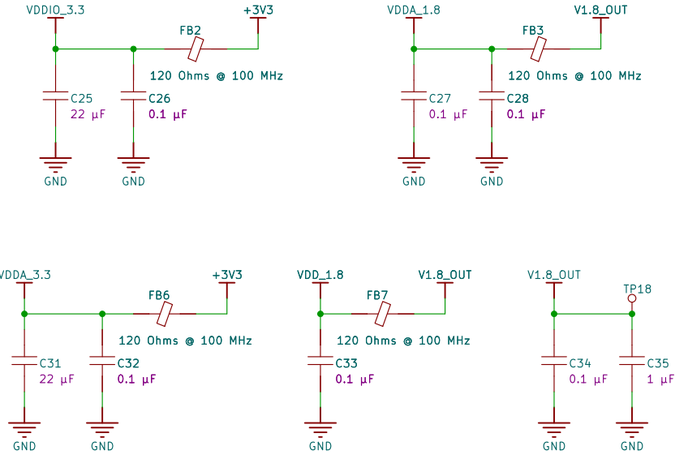
The impedance profiles will only be accurate if the following aspects are specified when ordering:

Layers = 6
Inner copper weight = 0.5oz
Outer copper weight = 1oz
PCB thickness = 1.6mm
Manufacturer = JLCPCB
Stack-up Code = JLC0616H-3313A

Analog Interface

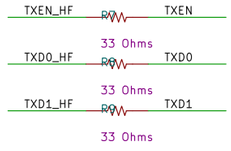


Transceiver Supply Decoupling

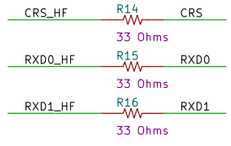


Slew Rate Control

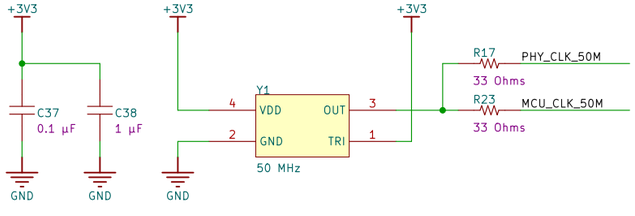
Place these sereis resistors close to the MCU



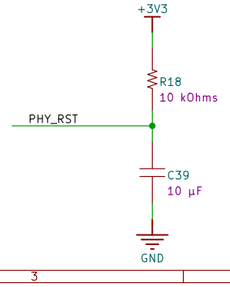
Place these series resistors close to the PHY



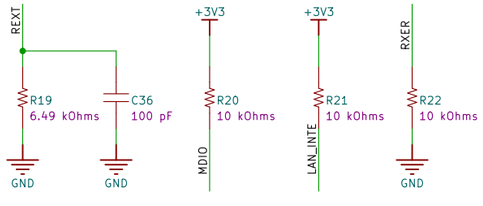
50 MHz Clock

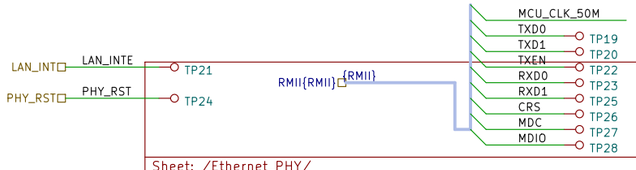


Reset



Biasing



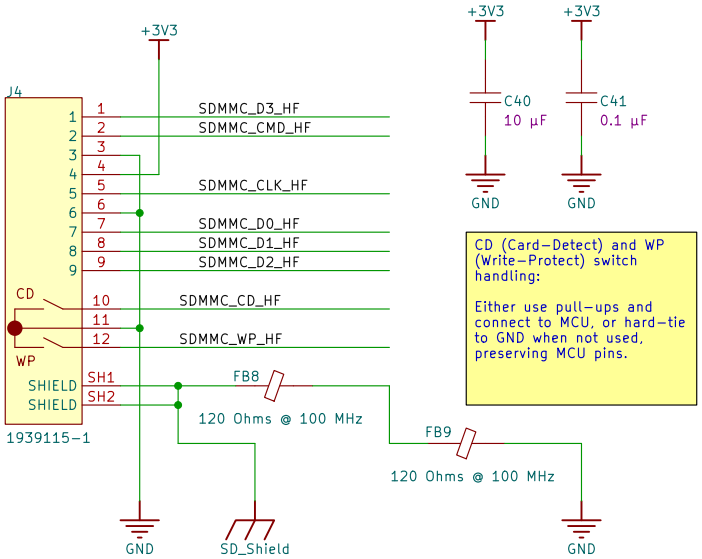


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Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 3/16

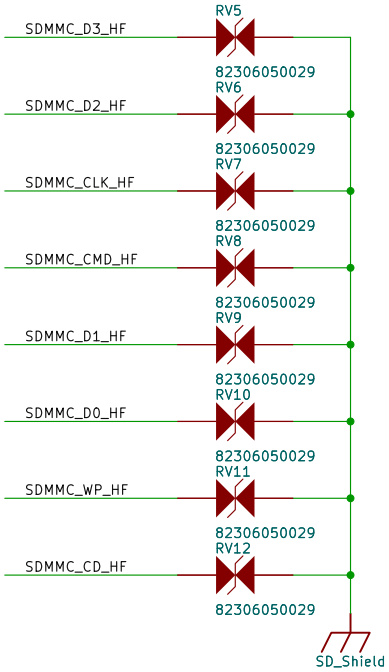
SD Card Slot



CD (Card-Detect) and WP (Write-Protect) switch handling:

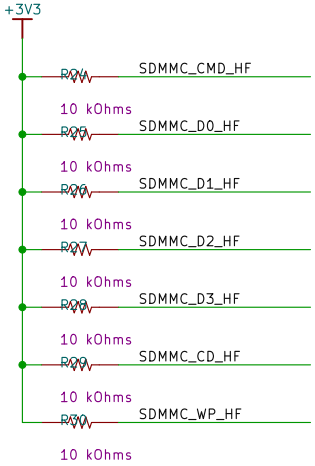
Either use pull-ups and connect to MCU, or hard-tie to GND when not used, preserving MCU pins.

ESD Protection



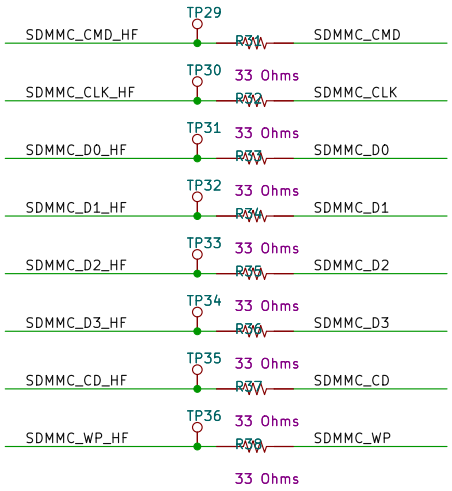
Pull-Up

Place these pullup resistors close to the MCU.



Slew Rate Control

Place these series resitors close to the MCU.





Layout Notes

Keep SDMMC traces below 120mm.

Avoid using vias on SDMMC signals, if necessary, use the same number of vias for all signals.

Length match the following signals to <10 mm:

CMD, CLK, D0, D1, D2, D3, length matched to each other (SDMMC net class).

Create a ground plane beneath the SD card connector with the SD_Shield net.

Terminate ESD protection diodes to SD_Shield plane.

Bond SD_Shield to GND using ferrite beads at a single bonding point.

The followoing signals need to be routed using a 50 Ohm coplanar microstrip or stripline impedance profile:

CMD, CLK, D0, D1, D2, D3.

The calculations for trace width have been made according to the manufacturing specs listed in the main architecture sheet of this PCB and assuming a coplanar routing approach (0.1mm clearance to GND).

Note that the signal layers of this PCB are Layer 1, 4 and 6 and the reference layers are Layer 2 and 5. If this is changed, the trace width has to be recalculated.

The results of the JLCPCB online impedance caluclator are the following:

Layer 1 trace width = 0.1199 mm
Layer 4 trace width = 0.2149 mm
Layer 6 trace width = 0.1199 mm

The impedance profiles will only be accurate if the following aspects are specified when ordering:

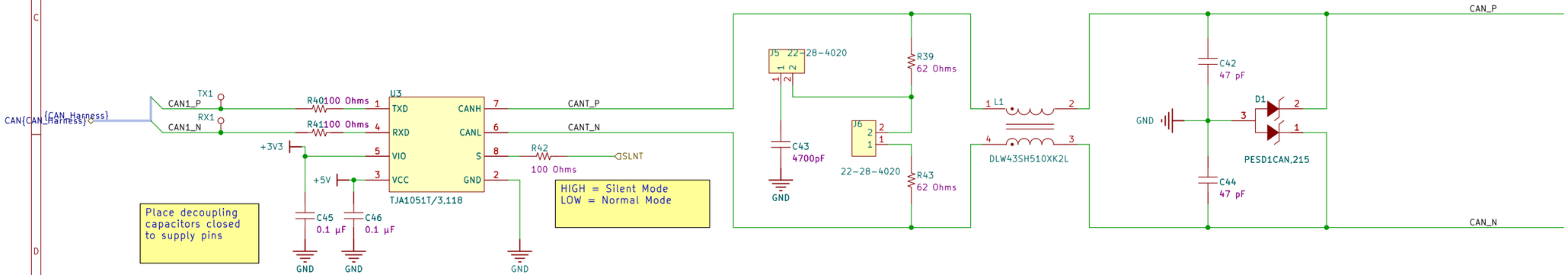
Layers = 6
Inner copper weight = 0.5oz
Outer copper weight = 1oz
PCB thickness = 1.6mm
Manufacturer = JLCPCB
Stack-up Code = JLC0616H-3313A

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File: SD.kicad_sch

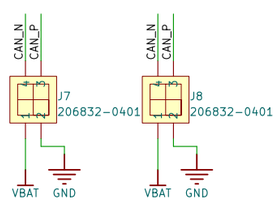
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 4/16

CAN

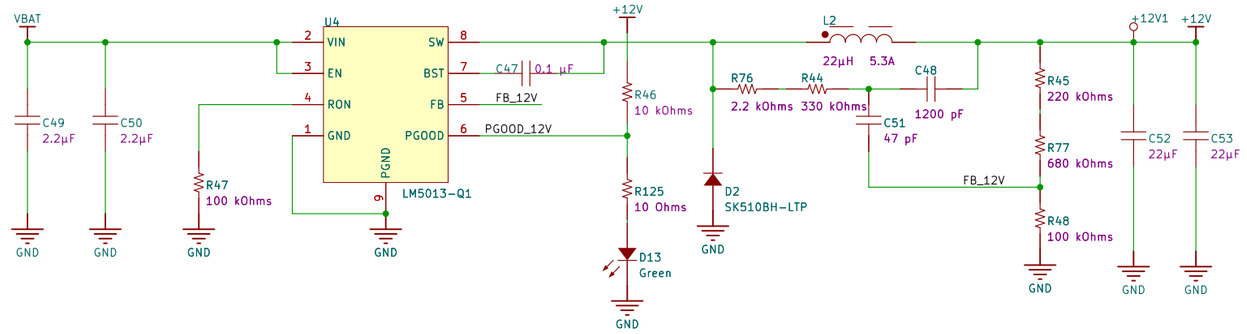


CAN Connector

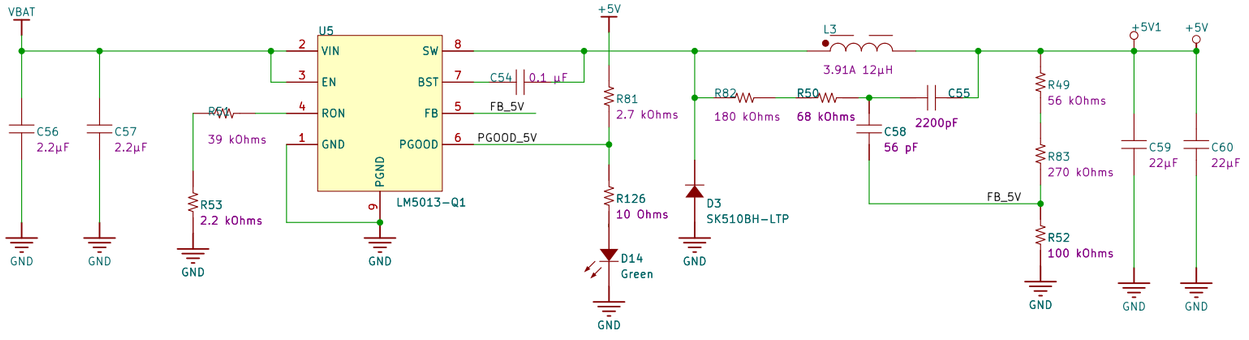


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KiCad E.D.A. 10.0.6		Id: 5/16

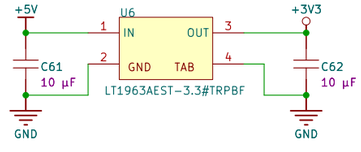
+12V Power Supply



+5V Power Supply

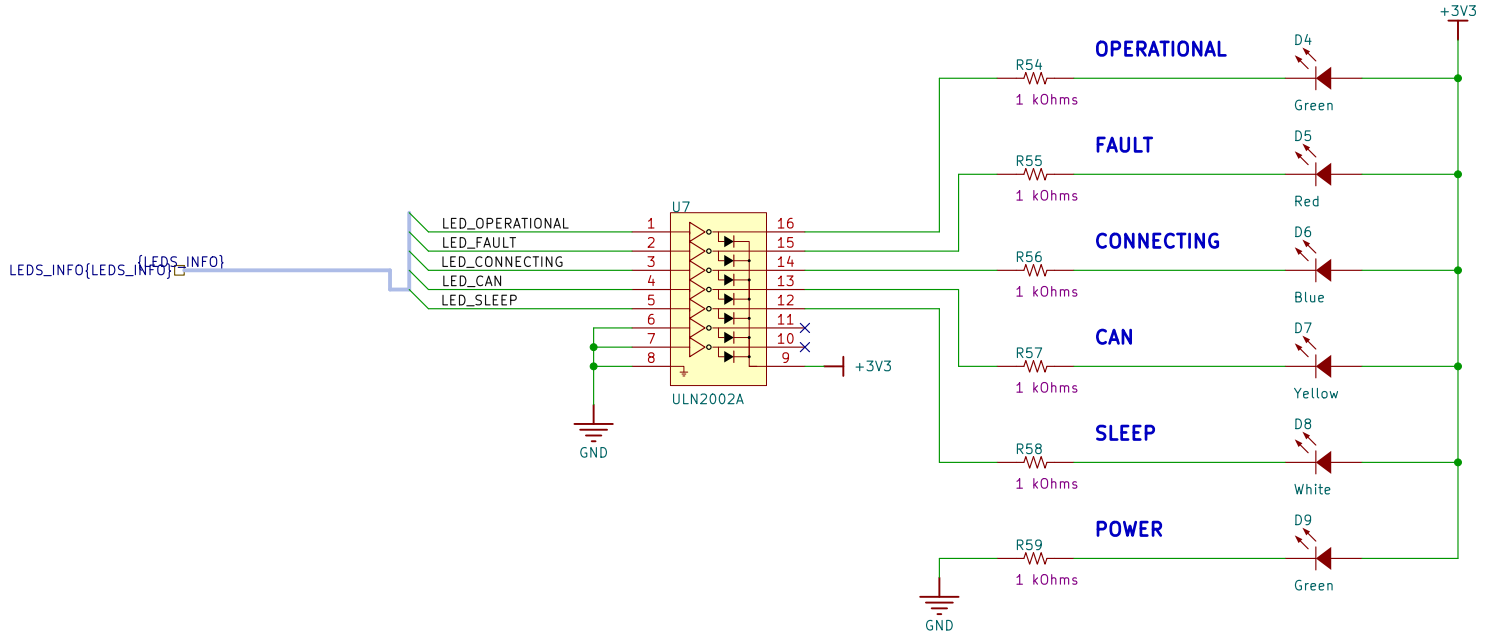


+3V3 Power Supply



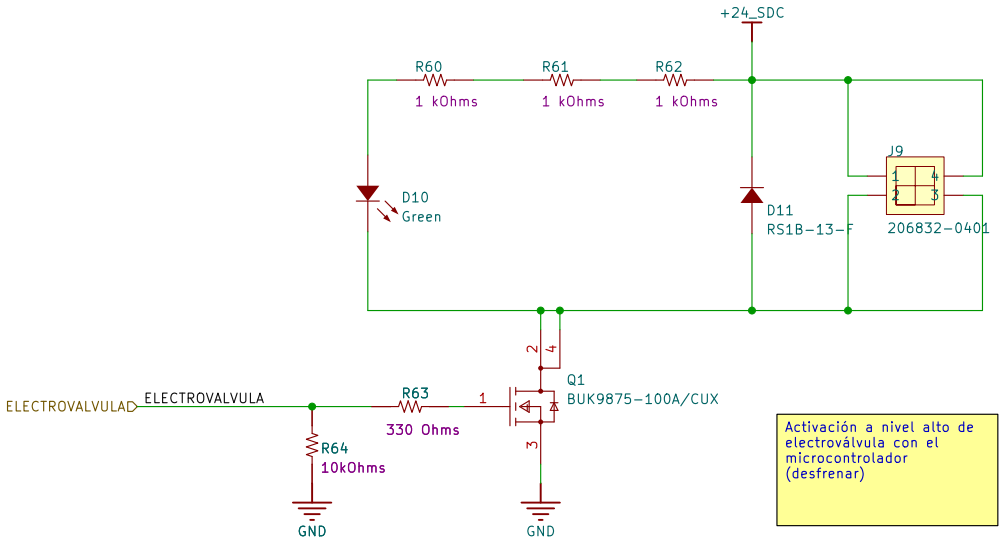
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Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 6/16

Information LEDs



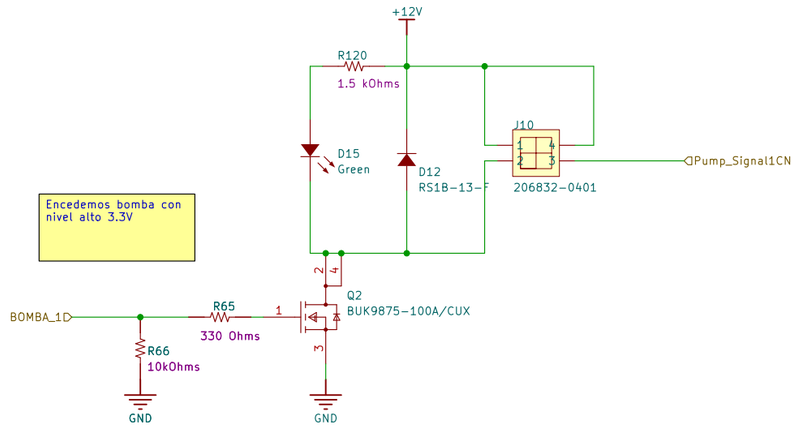
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 7/16

ELECTROVALVULA

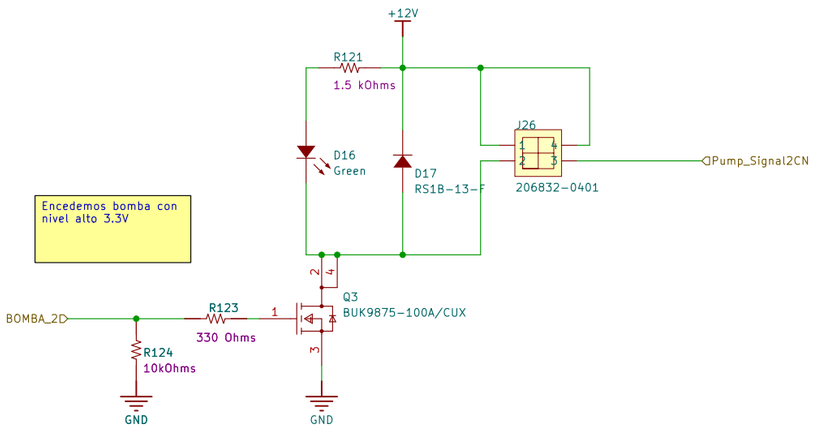


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KiCad E.D.A. 10.0.6		Id: 8/16

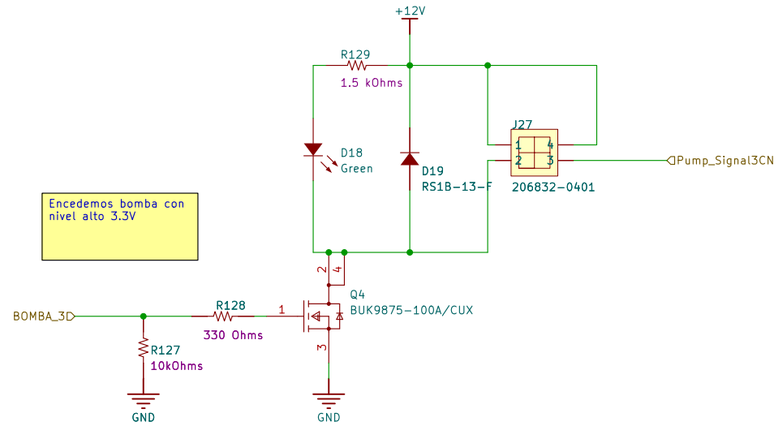
BOMBA_1



BOMBA_2



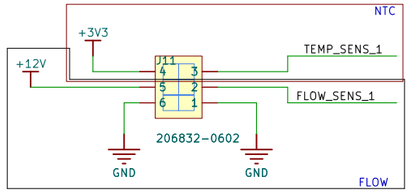
BOMBA_3



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Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 9/16

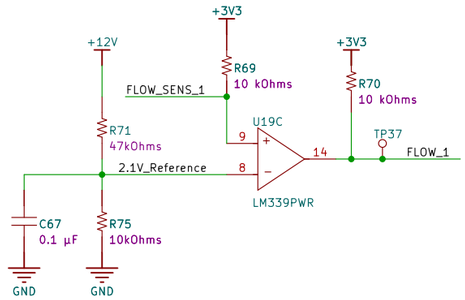
Caudalimetro_1

Conector_1

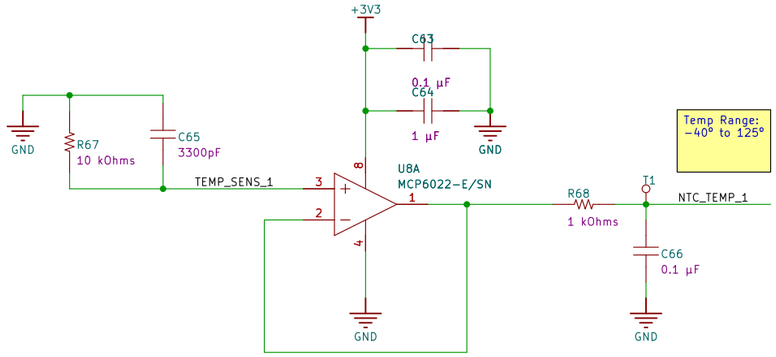


Como no sabemos si la salida digital del sensor de caudal es open drain o push pull, no nos podemos arriesgar a dañar el micro, por eso usamos este comparador, la idea es cuando se sepa en caso de que sea push pull no soldaremos la pull-up que se encuentra en el pin 5 (entrada positiva) y en caso de ser open drain soldaremos el comparador junto a la pull-up

Sensor_Caudal_1



Sensor_NTC_1

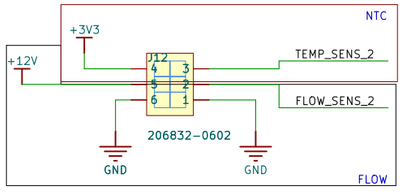


Adjunto aqui el link de la pagina web, en ella encontramos el datasheet y el manual: https://shop.alphacool.com/en/shop/controllers-and-sensors

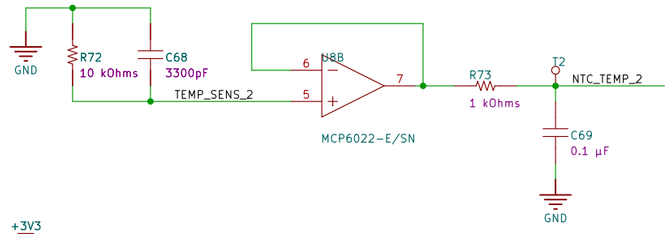
Explicando un poco este circuito del caudalimeto tenemos por un lado, un sensor de temp, el cual es una NTC que lee la temperatura del agua de la refrigeracion, y por otro lado tenemos un sensor de caudal, el cual es un sensor Hall con Tachometer/RPM Signal, esto indica que es salida digital Open Drain (por eso la pull-up), ademas este salida del sensor la deberemos de conector a un INPUT CAPTURE de la STM32, debido a que el codigo de la STM32 no mediremos voltaje, si no que el micro contará los pulsos que le llegan en periodo determinado para calcular la frecuencia (RPM)

Caudalimetro_2

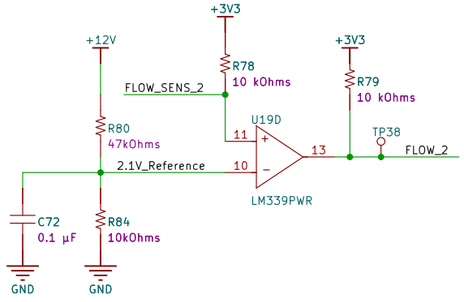
Conector_2

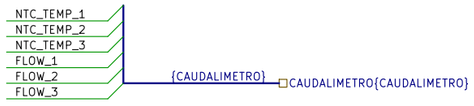


Sensor_NTC_2



Sensor_Caudal_2

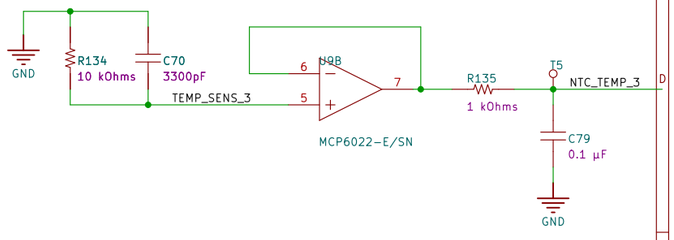




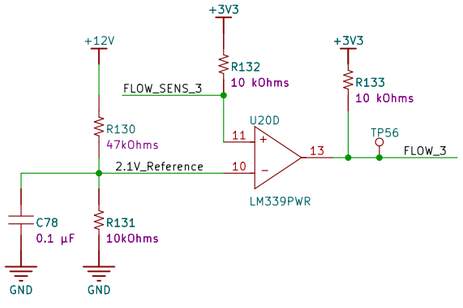
Caudalimetro_3

Conector_3

Sensor_NTC_3

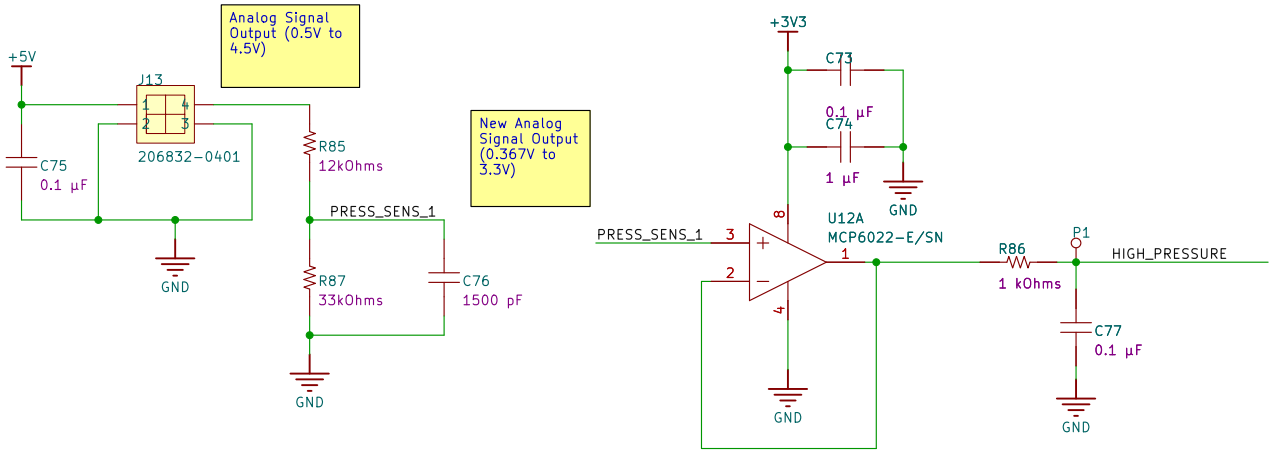


Sensor_Caudal_3



Sheet: /Caudalimetro/		
File: Caudalimetro.kicad_sch		
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Size: A3	Date:	**Rev:**
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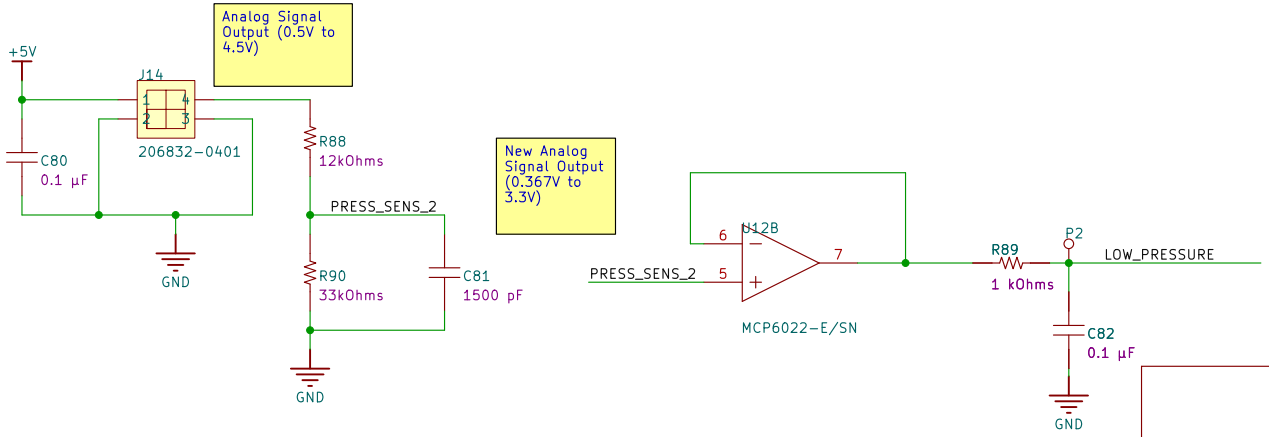
Transductor_Alta_Presion



Pin Out: Red = Vdc (5V) 10% to 90% Vs
Black = GND
Green =No se usa
White = Analog Signal Output (0.5V to 4.5V)

Ratiometrico
Rango de presion (0-5000 Psi)

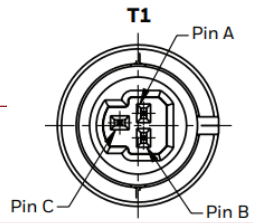
Transductor_Baja_Presion



Pin Out:
Pin A = Vdc (5V) 10% to 90% Vs
Pin B = GND
Pin C = Analog Signal Output (0.5V to 4.5V)

Ratiometrico
Rango de presion (0-10 Bar)

Como es Sealed Gage (parte de 1 atm = 1.013 Bar) no parte realmente de 0 a 10 Bar, si no que el rango de presion real (1.013-11.013 Bar)



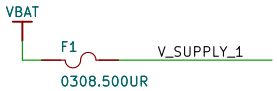
HIGH_PRESSURE
LOW_PRESSURE
{PRESSURE_SENS}
PRESSURE_SENS{PRESSURE_SENS}

Sheet: /Transductores_Presion/
File: Transductores_Presion.kicad_sch

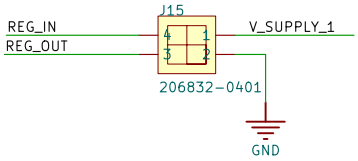
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 11/16

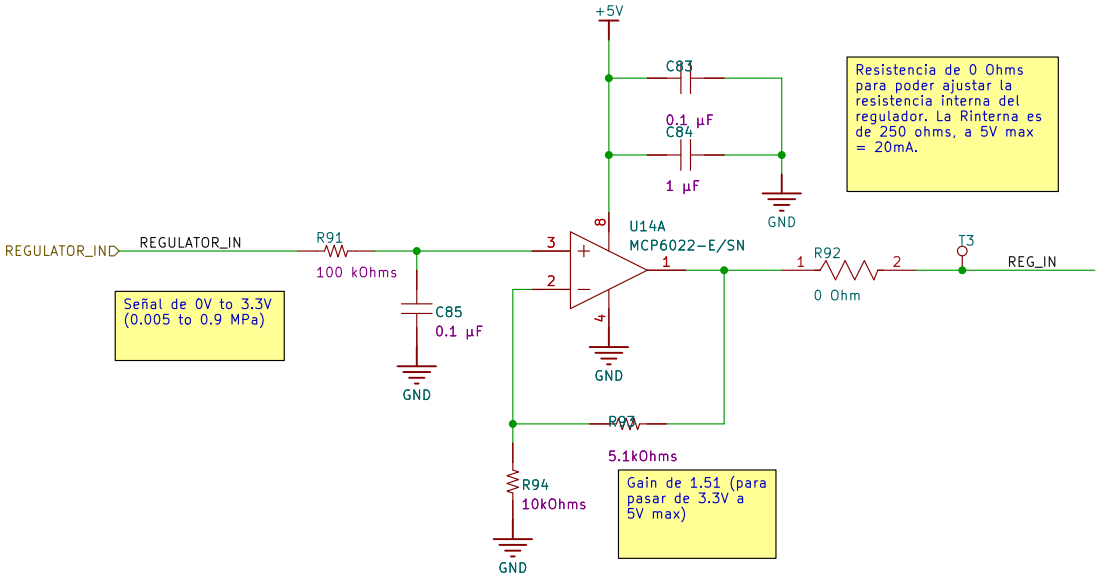
Supply



Conector



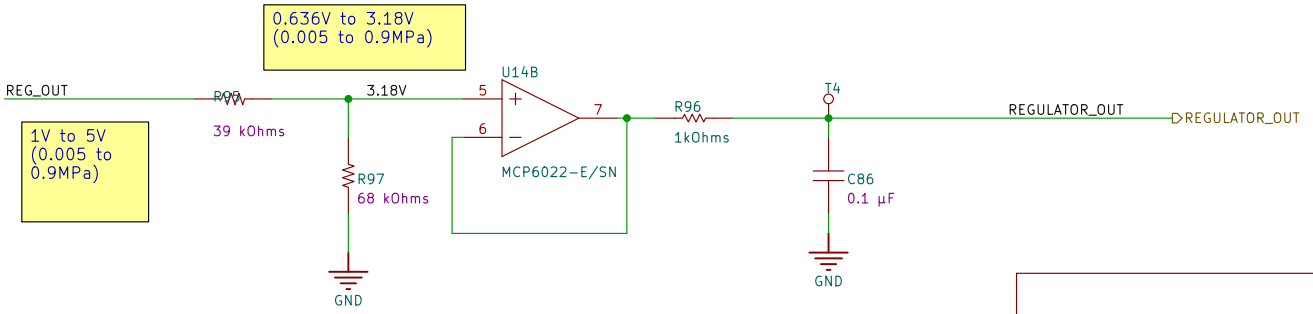
Regulator IN



Pin Out:
Brown = Vdc (24V)
Blue = GND
White = Analog Signal IN, sink type (4mA to 20mA)
Black = Analog Signal Monitoring (1V to 5V)

Rango de presion (0.005 to 0.9 MPa)

Regulator OUT



Sheet: /Regulador_Presion/		
File: Regulador_Presion.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 12/16

Limit switch logic

{SENSORS}

SENSORS{SENSORS}

XOR

AND

SENSOR_1_NO_OUT
SENSOR_1_NC_OUT
TP39
TP40
IC5A SN74LVC86AD

SENSOR_2_NO_OUT
SENSOR_2_NC_OUT
TP41
TP42

SENSOR_3_NO_OUT
SENSOR_3_NC_OUT
TP43
TP44

SENSOR_4_NO_OUT
SENSOR_4_NC_OUT
TP45
TP46

IC6A SN74LVC86AD

SENSOR_5_NO_OUT
SENSOR_5_NC_OUT
TP47
TP48

SENSOR_6_NO_OUT
SENSOR_6_NC_OUT
TP49
TP50

SENSOR_7_NO_OUT
SENSOR_7_NC_OUT
TP51
TP52

SENSOR_8_NO_OUT
SENSOR_8_NC_OUT
TP53
TP54

IC4A SN74LVC08AD
IC4B SN74LVC08AD
IC4C SN74LVC08AD
IC4D SN74LVC08AD
IC3A SN74LVC08AD
IC3B SN74LVC08AD
IC3C SN74LVC08AD
IC3D SN74LVC08AD

TP57 TP58 TP59 TP60 TP61 TP62 TP63 TP64 TP65 TP66 TP67 TP68 TP69 TP70 TP71

We need 3V3 in the gate driver RESET pin to enable PWMs

The RESET pin in the gate driver works like a ENABLE

NEVER solder both 0 ohm resistors at the same time

R104 0 Ohms GD_Reset_bypass
RESET
BRAKE_RESET
R107 0 Ohms
TP71 GD_Reset_bypass

J19 206832-0401
GND

Connector to PPU and LPU Gate Drivers

BRAKE_FAULT
BRAKE_FAULT

If BRAKE_FAULT = 0 something goes wrong

+3V3

IC3E SN74LVC08AD
IC4E SN74LVC08AD
IC5B SN74LVC86AD
IC7E SN74LVC08AD
IC8E SN74LVC08AD
IC6B SN74LVC86AD

C89 0.1uF
C90 0.1uF
C94 0.1uF
C93 0.1uF
C97 0.1uF
C100 0.1uF
VCC
GND

IC7A SN74LVC08AD
IC7B SN74LVC08AD
IC7C SN74LVC08AD
IC7D SN74LVC08AD
IC8A SN74LVC08AD
IC8B SN74LVC08AD
IC8C SN74LVC08AD
IC8D SN74LVC08AD

SENSOR_1_NO_OUT
SENSOR_2_NO_OUT
SENSOR_3_NO_OUT
SENSOR_4_NO_OUT
SENSOR_5_NO_OUT
SENSOR_6_NO_OUT
SENSOR_7_NO_OUT
SENSOR_8_NO_OUT

TP72 TP73 TP74 TP75 TP76 TP77 TP78 TP79

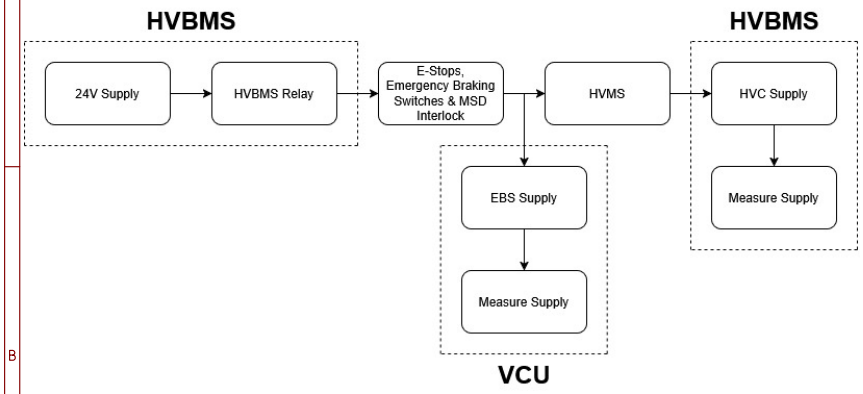
BRAKE_FAULT

RELEASED

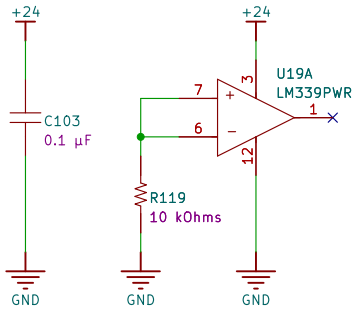
When Brake Disengaged, RELEASED = 1

Sheet: /Braking_Sensor/		
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Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 13/16

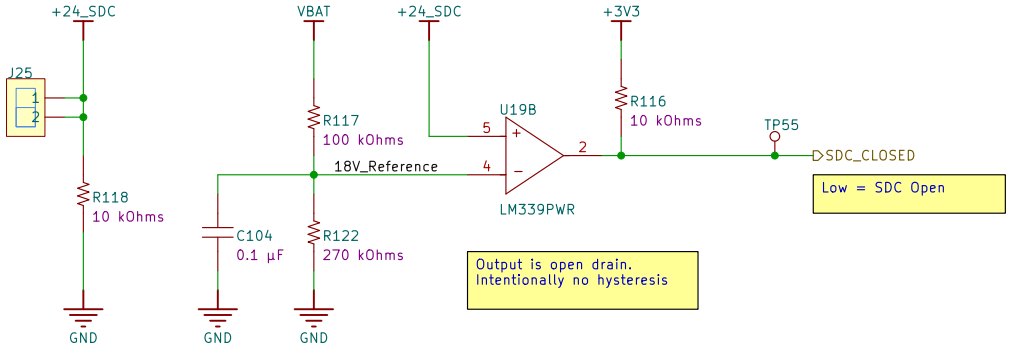
Shut-down Circuit (SDC)



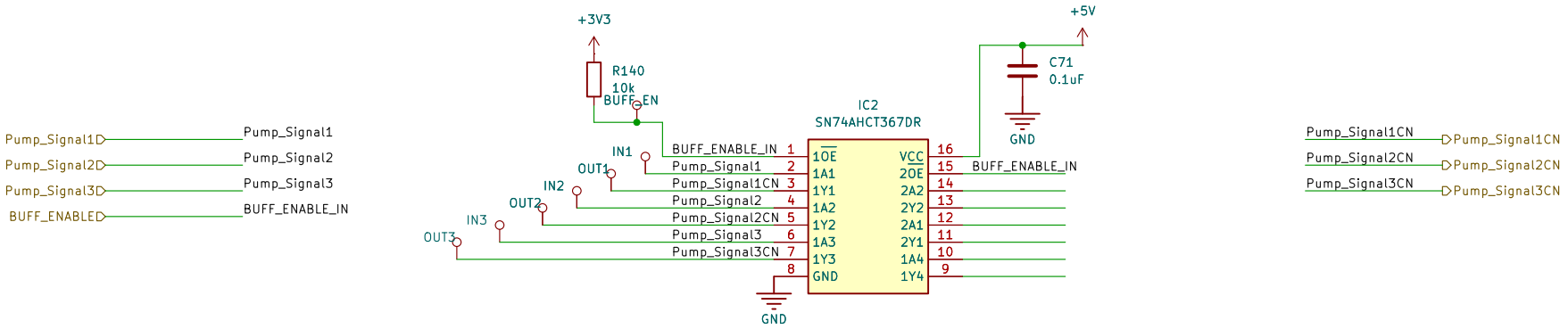
Decoupling & Unused Channels



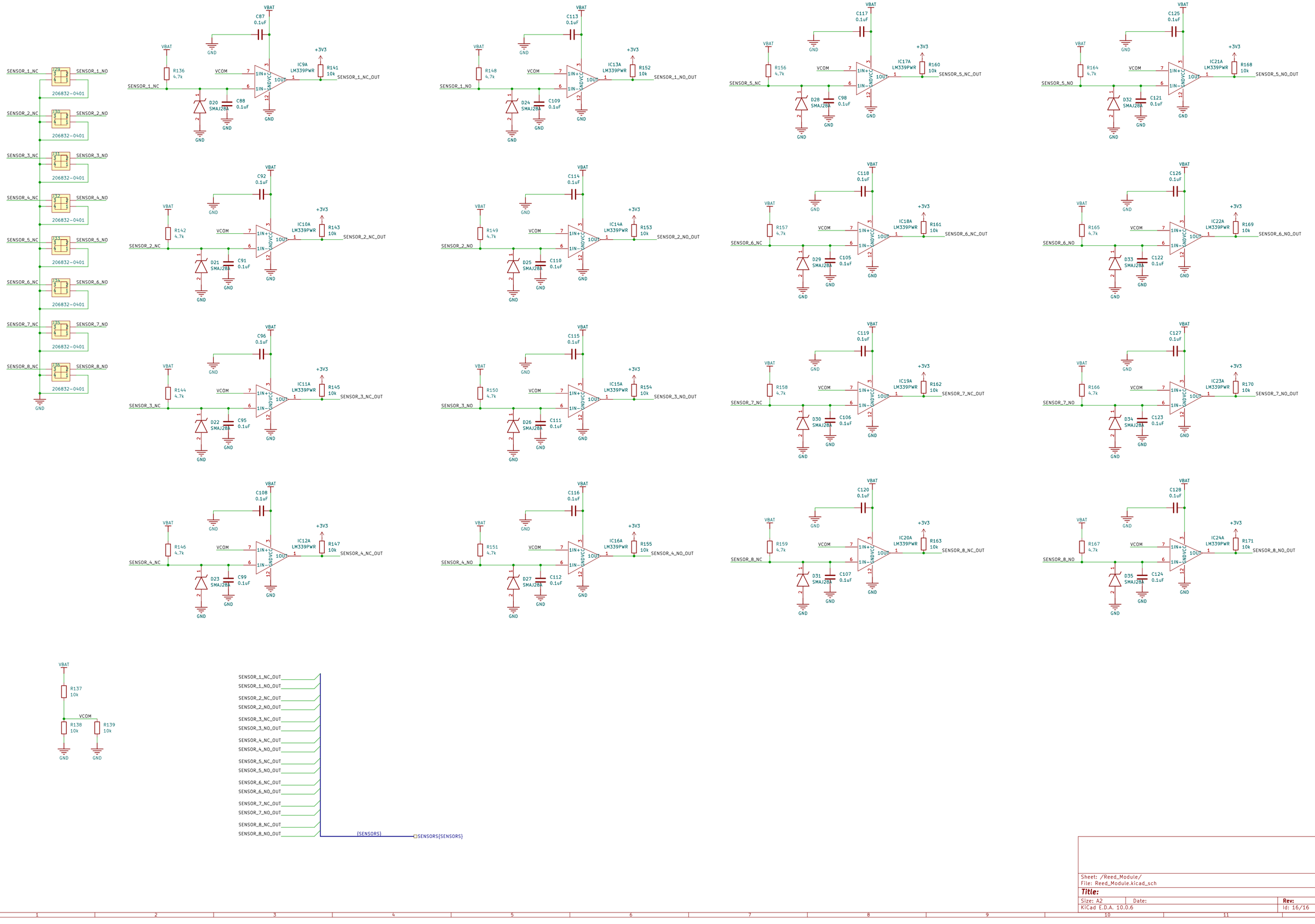
SDC Input & Measurement



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KiCad E.D.A. 10.0.6		Id: 14/16



Sheet: /Buffer/		
File: Buffer.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 15/16



Sheet: /Reed_Module/	
File: Reed_Module.kicad_sch	
Title:	
Size: A2	Date:
KiCad E.D.A. 10.0.6	**Rev:** Id: 16/16